

LECTURE 11

LINEAR PROCESSES III: ASYMPTOTIC RESULTS

(Phillips and Solo (1992) and Phillips' Lecture Notes on Stationary and Nonstationary Time Series)

In this lecture, we discuss the LLN and CLT for a linear process $\{X_t\}$ generated as

$$\begin{aligned} X_t &= \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} \\ &= C(L) \varepsilon_t, \end{aligned} \tag{1}$$

where

$$C(L) = \sum_{j=0}^{\infty} c_j L^j,$$

and $\{\varepsilon_t\}$ is a sequence of i.i.d. random variables with zero mean and finite variance. The LLN and CLT for $\{X_t\}$ rely only on the LLN and CLT for i.i.d. sequences and a certain decomposition of the lag polynomial $C(L)$.

The method also works for more general sequences with the ε_t being independent but not identically distributed (inid) or martingale difference sequences (mds).

Definition 1 Let $\{\mathcal{F}_t\}$ be an increasing sequence of σ -fields. Then $\{(u_t, \mathcal{F}_t)\}$ is a martingale if

$$\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = u_{t-1}$$

with probability one. The sequence $\{(\varepsilon_t, \mathcal{F}_t)\}$ is said to be a martingale difference sequence if

$$\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$$

with probability one.

For $\{u_t\}$, the σ -field $\mathcal{F}_t$ is often taken to be $\sigma(u_t, u_{t-1}, \dots)$. Suppose that $\{(u_t, \mathcal{F}_t)\}$ is a martingale. Then $\{(u_t - u_{t-1}, \mathcal{F}_t)\}$ is an mds, since $u_t - u_{t-1} = u_t - \mathbb{E}[u_t \mid \mathcal{F}_{t-1}]$, and, therefore, $\mathbb{E}[u_t - u_{t-1} \mid \mathcal{F}_{t-1}] = 0$. Each of the conditions that $\{\varepsilon_t\}$ be i.i.d., inid, or an mds is stronger than white noise alone.

Lemma 1 (SLLN for inid sequences, White (2001), Corollary 3.9) Let $\{\varepsilon_t\}$ be a sequence of independent random variables such that $\sup_t \mathbb{E}[|\varepsilon_t|^{1+\delta}] < \infty$ for some $\delta > 0$. Then, $n^{-1} \sum_{t=1}^n \varepsilon_t - n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t] \rightarrow_{a.s.} 0$.

Lemma 2 (SLLN for mds, White (2001), Exercise 3.77) Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be an mds such that $\sup_t \mathbb{E}[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$. Then, $n^{-1} \sum_{t=1}^n \varepsilon_t \rightarrow_{a.s.} 0$.

Lemma 3 (CLT for inid sequences, White (2001), Theorem 5.10) Let $\{\varepsilon_t\}$ be a sequence of independent random variables such that $\mathbb{E}[\varepsilon_t] = 0$ for all t , $\sup_t \mathbb{E}[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$, and for all n sufficiently large $n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t^2] > \delta_0 > 0$. Then, $n^{-1/2} \sum_{t=1}^n \varepsilon_t / (n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t^2])^{1/2} \rightarrow_d N(0, 1)$.

Lemma 4 (CLT for mds, White (2001), Corollary 5.26) Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be an mds such that $\sup_t \mathbb{E}[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$. Suppose that for all n sufficiently large $n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t^2] > \delta_0 > 0$, and $n^{-1} \sum_{t=1}^n \varepsilon_t^2 - n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t^2] \rightarrow_p 0$. Then, $n^{-1/2} \sum_{t=1}^n \varepsilon_t / (n^{-1} \sum_{t=1}^n \mathbb{E}[\varepsilon_t^2])^{1/2} \rightarrow_d N(0, 1)$.

Lemma 5 (CLT for strictly stationary and ergodic mds) Let $\{(\varepsilon_t, \mathcal{F}_t)\}$ be a strictly stationary and ergodic mds such that $\mathbb{E}[\varepsilon_t^2] < \infty$. Then $n^{-1/2} \sum_{t=1}^n \varepsilon_t \rightarrow_d N(0, \mathbb{E}[\varepsilon_t^2])$.

Beveridge and Nelson (BN) decomposition

First, we discuss an algebraic decomposition of a lag polynomial into long-run and transient components. The decomposition was introduced by Beveridge and Nelson (1981).

Lemma 6 *Let $C(L) = \sum_{j=0}^{\infty} c_j L^j$. Then*

- (a) $C(L) = C(1) - (1 - L)\tilde{C}(L)$, where $\tilde{C}(L) = \sum_{j=0}^{\infty} \tilde{c}_j L^j$ with $\tilde{c}_j = \sum_{h=j+1}^{\infty} c_h$.
- (b) If $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$, then $\sum_{j=0}^{\infty} \tilde{c}_j^2 < \infty$.
- (c) If $\sum_{j=1}^{\infty} j |c_j| < \infty$, then $\sum_{j=0}^{\infty} |\tilde{c}_j| < \infty$.

Proof. For part (a), write

$$\begin{aligned}
 \sum_{j=0}^{\infty} c_j L^j &= \sum_{j=0}^{\infty} c_j - \sum_{j=1}^{\infty} c_j \\
 &\quad + \left(\sum_{j=1}^{\infty} c_j - \sum_{j=2}^{\infty} c_j \right) L \\
 &\quad + \left(\sum_{j=2}^{\infty} c_j - \sum_{j=3}^{\infty} c_j \right) L^2 \\
 &\quad + \dots \\
 &\quad + \left(\sum_{j=h}^{\infty} c_j - \sum_{j=h+1}^{\infty} c_j \right) L^h \\
 &\quad + \dots
 \end{aligned}$$

Rearranging the terms,

$$\begin{aligned}
 \sum_{j=0}^{\infty} c_j L^j &= \sum_{j=0}^{\infty} c_j \\
 &\quad - (1 - L) \sum_{j=1}^{\infty} c_j \\
 &\quad - (1 - L) \sum_{j=2}^{\infty} c_j L \\
 &\quad - \dots \\
 &\quad - (1 - L) \sum_{j=h+1}^{\infty} c_j L^h \\
 &\quad - \dots \\
 &= \sum_{j=0}^{\infty} c_j - (1 - L) \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} c_h \right) L^j \\
 &= C(1) - (1 - L)\tilde{C}(L).
 \end{aligned}$$

For part (b),

$$\begin{aligned}
\sum_{j=0}^{\infty} \tilde{c}_j^2 &= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} c_h \right)^2 \\
&\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h| \right)^2 \\
&= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h|^{1/2} h^{1/4} |c_h|^{1/2} h^{-1/4} \right)^2 \\
&\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} |c_h| h^{1/2} \right) \left(\sum_{h=j+1}^{\infty} |c_h| h^{-1/2} \right) \\
&\leq \left(\sum_{h=0}^{\infty} |c_h| h^{1/2} \right) \sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2}.
\end{aligned}$$

Next, consider $\sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2}$. The term $|c_1|$ appears in the sum only once, when $j = 0$. The term $|c_2|$ appears in the sum twice, when $j = 0, 1$. Hence, $|c_h|$ appears when $j = 0, 1, \dots, h-1$, a total of h times. Therefore,

$$\begin{aligned}
\sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| h^{-1/2} &= \sum_{j=1}^{\infty} |c_j| j^{-1/2} j \\
&= \sum_{j=1}^{\infty} |c_j| j^{1/2},
\end{aligned}$$

and

$$\sum_{j=0}^{\infty} \tilde{c}_j^2 \leq \left(\sum_{j=0}^{\infty} |c_j| j^{1/2} \right)^2.$$

For part (c),

$$\begin{aligned}
\sum_{j=0}^{\infty} |\tilde{c}_j| &= \sum_{j=0}^{\infty} \left| \sum_{h=j+1}^{\infty} c_h \right| \\
&\leq \sum_{j=0}^{\infty} \sum_{h=j+1}^{\infty} |c_h| \\
&= \sum_{j=0}^{\infty} |c_j| j.
\end{aligned}$$

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The assumptions $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ and $\sum_{j=1}^{\infty} j |c_j| < \infty$ are stronger than absolute summability $\sum_{j=1}^{\infty} |c_j| < \infty$, which ensures finiteness of the long-run variance.

According to the BN decomposition, if $\{X_t\}$ is a linear process, then

$$\begin{aligned}
X_t &= C(L)\varepsilon_t \\
&= C(1)\varepsilon_t - (1-L)\tilde{C}(L)\varepsilon_t \\
&= C(1)\varepsilon_t - (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}),
\end{aligned} \tag{2}$$

where

$$\tilde{\varepsilon}_t = \tilde{C}(L)\varepsilon_t.$$

Furthermore, $\tilde{\varepsilon}_t$ has finite variance provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$. The first summand on the right-hand side of (2), $C(1)\varepsilon_t$, is the long-run component, and $\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}$ is transient.

A similar decomposition exists in the vector case. The transient component has finite variance if $\sum_{j=1}^{\infty} j^{1/2} \|C_j\| < \infty$:

$$\begin{aligned} \sum_{j=0}^{\infty} \|\tilde{C}_j\|^2 &= \sum_{j=0}^{\infty} \left\| \sum_{h=j+1}^{\infty} C_h \right\|^2 \\ &\leq \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} \|C_h\| \right)^2 \\ &= \sum_{j=0}^{\infty} \left(\sum_{h=j+1}^{\infty} \|C_h\|^{1/2} h^{1/4} \|C_h\|^{1/2} h^{-1/4} \right)^2 \\ &\leq \left(\sum_{j=0}^{\infty} j^{1/2} \|C_j\| \right)^2. \end{aligned}$$

The condition in part (c) of Lemma 6 becomes $\sum_{j=1}^{\infty} j \|C_j\| < \infty$ in the vector case.

WLLN

Suppose that

- $X_t = C(L)\varepsilon_t$.
- $\{\varepsilon_t\}$ is a sequence of i.i.d. random variables with $E[|\varepsilon_t|] < \infty$ and $E[\varepsilon_t] = 0$.
- $C(L)$ satisfies $\sum_{j=1}^{\infty} j |c_j| < \infty$.

We will show that under these conditions

$$n^{-1} \sum_{t=1}^n X_t \rightarrow_p 0.$$

The key is the BN decomposition (2). The sum $\sum_{t=1}^n (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1})$ is a telescoping sum:

$$\begin{aligned} \sum_{t=1}^n (\tilde{\varepsilon}_t - \tilde{\varepsilon}_{t-1}) &= (\tilde{\varepsilon}_1 - \tilde{\varepsilon}_0) + (\tilde{\varepsilon}_2 - \tilde{\varepsilon}_1) + (\tilde{\varepsilon}_3 - \tilde{\varepsilon}_2) + \dots + (\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}) \\ &= \tilde{\varepsilon}_n - \tilde{\varepsilon}_0, \end{aligned}$$

so that

$$n^{-1} \sum_{t=1}^n X_t = C(1)n^{-1} \sum_{t=1}^n \varepsilon_t - n^{-1} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0).$$

Since $\sum_{j=1}^{\infty} j |c_j| < \infty$, $|C(1)| < \infty$. Hence, by the WLLN for i.i.d. sequences,

$$C(1)n^{-1} \sum_{t=1}^n \varepsilon_t \rightarrow_p 0.$$

Next,

$$\Pr(n^{-1} |\tilde{\varepsilon}_t| > \delta) \leq \frac{\mathbb{E}[|\tilde{\varepsilon}_t|]}{n\delta},$$

and

$$\begin{aligned} \mathbb{E}[|\tilde{\varepsilon}_t|] &= \mathbb{E}\left[\left|\sum_{j=0}^{\infty} \tilde{c}_j \varepsilon_{t-j}\right|\right] \\ &\leq \sum_{j=0}^{\infty} |\tilde{c}_j| \mathbb{E}[|\varepsilon_{t-j}|] \\ &= \mathbb{E}[|\varepsilon_0|] \sum_{j=0}^{\infty} |\tilde{c}_j| \\ &< \infty, \end{aligned}$$

provided that $\sum_{j=1}^{\infty} j |c_j| < \infty$. Therefore,

$$n^{-1} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0) \rightarrow_p 0.$$

If we assume that $\mathbb{E}[\varepsilon_t^2] < \infty$, then we can replace $\sum_{j=1}^{\infty} j |c_j| < \infty$ with $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$, since

$$\Pr(n^{-1} |\tilde{\varepsilon}_t| > \delta) \leq \frac{\mathbb{E}[\tilde{\varepsilon}_t^2]}{n^2 \delta^2},$$

and $\mathbb{E}[\tilde{\varepsilon}_t^2] < \infty$ provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ holds.

We can prove similar WLLNs with $\{\varepsilon_t\}$ being inid or an mds by using the corresponding LLNs for inid or mds. For example, the result holds if $\{\varepsilon_t\}$ is inid, $\sup_t \mathbb{E}[|\varepsilon_t|^{1+\delta}] < \infty$ for some $\delta > 0$, and $\sum_{j=1}^{\infty} j |c_j| < \infty$.

If $\{(\varepsilon_t, \mathcal{F}_t)\}$ is an mds, the result holds with $\sup_t \mathbb{E}[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$, and $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.

The preceding results assumed that $\mathbb{E}[X_t] = 0$. To allow for a nonzero mean, replace the first assumption with $X_t = \mu + C(L)\varepsilon_t$, where μ is the mean of X_t . In this case, under the same set of conditions, we have $n^{-1} \sum_{t=1}^n X_t \rightarrow_p \mu$. For example, an AR(1) process with mean μ is given by $(1 - \phi L)(X_t - \mu) = \varepsilon_t$. If $|\phi| < 1$, then the sample average of X_t converges in probability to μ provided that the corresponding moment restrictions hold.

CLT

Suppose that

- $X_t = C(L)\varepsilon_t$.
- $\{\varepsilon_t\}$ is a sequence of i.i.d. random variables with $\mathbb{E}[\varepsilon_t^2] = \sigma^2 < \infty$ and $\mathbb{E}[\varepsilon_t] = 0$.
- $C(L)$ satisfies $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.
- $C(1) \neq 0$.

The BN decomposition allows us to write

$$\begin{aligned} n^{-1/2} \sum_{t=1}^n X_t &= C(1)n^{-1/2} \sum_{t=1}^n \varepsilon_t - n^{-1/2} (\tilde{\varepsilon}_n - \tilde{\varepsilon}_0) \\ &= C(1)n^{-1/2} \sum_{t=1}^n \varepsilon_t + o_p(1) \\ &\rightarrow_d C(1)N(0, \sigma^2) \\ &= N(0, \sigma^2 C(1)^2). \end{aligned}$$

Here, convergence in distribution is by the CLT for i.i.d. random variables. The approach illustrates why, in the serially correlated case, the asymptotic variance depends on the long-run variance of $\{X_t\}$. Again, the approach can be extended to the case where $\{\varepsilon_t\}$ is inid or mds.

In the vector case, suppose that $\{\varepsilon_t\}$ is a sequence of i.i.d. k -vectors with $E[\varepsilon_t] = 0$, and $E[\varepsilon_t \varepsilon_t^\top] = \Sigma$, a finite matrix. Let $X_t = C(L)\varepsilon_t$, and $\sum_{j=1}^{\infty} j^{1/2} \|C_j\| < \infty$, $C(1) \neq 0$. Since $n^{-1/2} \sum_{t=1}^n \varepsilon_t \rightarrow_d N(0, \Sigma)$, we have that

$$n^{-1/2} \sum_{t=1}^n X_t \rightarrow_d N(0, C(1)\Sigma C(1)^\top).$$

Convergence of sample variances

Estimators of coefficients in the linear regression model involve second sample moments $n^{-1} \sum_{t=1}^n X_t X_t^\top$. Here, we discuss convergence of sample second moments when $\{X_t\}$ is a linear process. We assume that $\{X_t\}$ is a scalar linear process satisfying the same assumptions as in the previous section. Write

$$\begin{aligned} X_t^2 &= (C(L)\varepsilon_t)^2 \\ &= \left(\sum_{j=0}^{\infty} c_j \varepsilon_{t-j} \right)^2 \\ &= \sum_{j=0}^{\infty} \sum_{l=0}^{\infty} c_j c_l \varepsilon_{t-j} \varepsilon_{t-l} \\ &= \sum_{j=0}^{\infty} c_j^2 \varepsilon_{t-j}^2 + 2 \sum_{j=0}^{\infty} \sum_{l>j}^{\infty} c_j c_l \varepsilon_{t-j} \varepsilon_{t-l} \\ &= \sum_{j=0}^{\infty} c_j^2 \varepsilon_{t-j}^2 + 2 \sum_{h=1}^{\infty} \sum_{j=0}^{\infty} c_j c_{j+h} \varepsilon_{t-j} \varepsilon_{t-j-h} \quad (\text{change of variable } l = j + h, \text{ so that } h = 1, 2, \dots) \\ &= B_0(L)\varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(L)\varepsilon_t \varepsilon_{t-h}, \end{aligned}$$

where for $h = 0, 1, \dots$,

$$\begin{aligned} B_h(L) &= \sum_{j=0}^{\infty} b_{h,j} L^j \\ &= \sum_{j=0}^{\infty} c_j c_{j+h} L^j. \end{aligned}$$

Thus,

$$\begin{aligned} B_0(L) &= \sum_{j=0}^{\infty} b_{0,j} L^j = \sum_{j=0}^{\infty} c_j^2 L^j, \\ B_1(L) &= \sum_{j=0}^{\infty} b_{1,j} L^j = \sum_{j=0}^{\infty} c_j c_{j+1} L^j, \\ &\dots \end{aligned}$$

The BN decomposition of $B_h(L)$ is

$$B_h(L) = B_h(1) - (1 - L)\tilde{B}_h(L), \tag{3}$$

where

$$\begin{aligned}\tilde{B}_h(L) &= \sum_{j=0}^{\infty} \tilde{b}_{h,j} L^j, \\ \tilde{b}_{h,j} &= \sum_{l=j+1}^{\infty} b_{h,l} \\ &= \sum_{l=j+1}^{\infty} c_l c_{l+h}.\end{aligned}$$

The BN decomposition of $B_h(L)$ is valid provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$:

$$\begin{aligned}\sum_{j=0}^{\infty} \tilde{b}_{h,j}^2 &= \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} c_l c_{l+h} \right)^2 \\ &= \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} l^{1/4} c_l c_{l+h} l^{-1/4} \right)^2 \\ &\leq \sum_{j=0}^{\infty} \left(\sum_{l=j+1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{l=j+1}^{\infty} c_{l+h}^2 l^{-1/2} \right) \\ &\leq \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{j=0}^{\infty} \sum_{l=j+1}^{\infty} c_{l+h}^2 l^{-1/2} \right) \\ &= \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right) \left(\sum_{l=0}^{\infty} c_{l+h}^2 l^{1/2} \right) \\ &\leq \left(\sum_{l=1}^{\infty} l^{1/2} c_l^2 \right)^2,\end{aligned}$$

and $\sum_{l=1}^{\infty} l^{1/2} c_l^2$ is finite provided that $\sum_{l=1}^{\infty} l^{1/2} |c_l|$ is finite:

$$\begin{aligned}\sum_{l=1}^{\infty} l^{1/2} c_l^2 &= \sum_{l=1}^{\infty} l^{1/2} |c_l|^2 \\ &\leq \sup_j |c_j| \sum_{l=1}^{\infty} l^{1/2} |c_l| < \infty,\end{aligned}$$

where $\sup_j |c_j| < \infty$ because $\sum_{l=1}^{\infty} l^{1/2} |c_l| < \infty$ and therefore $|c_l| \rightarrow 0$ as $l \rightarrow \infty$.

Thus, we have

$$\begin{aligned}
X_t^2 &= B_0(L)\varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(L)\varepsilon_t\varepsilon_{t-h} \\
&= B_0(1)\varepsilon_t^2 + 2 \sum_{h=1}^{\infty} B_h(1)\varepsilon_t\varepsilon_{t-h} \\
&\quad - (1-L) \left(\tilde{B}_0(L)\varepsilon_t^2 + 2 \sum_{h=1}^{\infty} \tilde{B}_h(L)\varepsilon_t\varepsilon_{t-h} \right) \\
&= \left(\sum_{j=0}^{\infty} c_j^2 \right) \varepsilon_t^2 + u_t - (1-L)\tilde{v}_t,
\end{aligned}$$

where

$$\begin{aligned}
u_t &= \varepsilon_t \left(2 \sum_{h=1}^{\infty} B_h(1)\varepsilon_{t-h} \right), \\
\tilde{v}_t &= \tilde{B}_0(L)\varepsilon_t^2 + 2 \sum_{h=1}^{\infty} \tilde{B}_h(L)\varepsilon_t\varepsilon_{t-h}.
\end{aligned}$$

We have

$$n^{-1} \sum_{t=1}^n X_t^2 = \left(\sum_{j=0}^{\infty} c_j^2 \right) n^{-1} \sum_{t=1}^n \varepsilon_t^2 + n^{-1} \sum_{t=1}^n u_t - n^{-1} (\tilde{v}_n - \tilde{v}_0).$$

We will show next that

$$n^{-1} \sum_{t=1}^n u_t \rightarrow_{a.s.} 0.$$

Let $\mathcal{F}_t = \sigma(\varepsilon_t, \varepsilon_{t-1}, \dots)$. Then $\{(u_t, \mathcal{F}_t)\}$ is an mds.

Lemma 7 (White (2001), Theorem 3.76) *Let $\{(u_t, \mathcal{F}_t)\}$ be an mds. If for some $r \geq 1$,*

$$\sum_{t=1}^{\infty} \mathbb{E} \left[|u_t|^{2r} \right] / t^{1+r} < \infty,$$

then $n^{-1} \sum_{t=1}^n u_t \rightarrow_{a.s.} 0$.

We will verify that $\{u_t\}$ satisfies the condition of the above lemma. Set $r = 1$. The condition is satisfied if $\sup_t \mathbb{E} [u_t^2] < \infty$, since $\sum_{t=1}^{\infty} t^{-2} < \infty$.

$$\begin{aligned}
\mathbb{E} [u_t^2] &= 4\sigma^2 \mathbb{E} \left[\left(\sum_{h=1}^{\infty} B_h(1)\varepsilon_{t-h} \right)^2 \right] \\
&= 4\sigma^4 \left(\sum_{h=1}^{\infty} B_h(1)^2 \right).
\end{aligned}$$

Next,

$$\begin{aligned}
\sum_{h=1}^{\infty} B_h(1)^2 &= \sum_{h=1}^{\infty} \left(\sum_{j=0}^{\infty} c_j c_{j+h} \right)^2 \\
&\leq \sum_{h=1}^{\infty} \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{j=0}^{\infty} c_{j+h}^2 \\
&= \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{h=1}^{\infty} \sum_{j=0}^{\infty} c_{j+h}^2 \\
&= \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{j=1}^{\infty} j c_j^2,
\end{aligned}$$

and, by the same argument as on page 3 of Lecture 10, $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$ implies that $\sum_{j=1}^{\infty} j c_j^2 < \infty$ as well.

By the same argument as in the WLLN section,

$$n^{-1} (\tilde{v}_n - \tilde{v}_0) \rightarrow_p 0,$$

provided that $\sum_{j=1}^{\infty} j^{1/2} |c_j| < \infty$.

Lastly, by the WLLN for i.i.d. sequences,

$$n^{-1} \sum_{t=1}^n \varepsilon_t^2 \rightarrow_p \sigma^2.$$

Therefore,

$$\begin{aligned}
n^{-1} \sum_{t=1}^n X_t^2 &\rightarrow_p \sigma^2 \sum_{j=0}^{\infty} c_j^2 \\
&= \mathbb{E} [X_t^2].
\end{aligned}$$